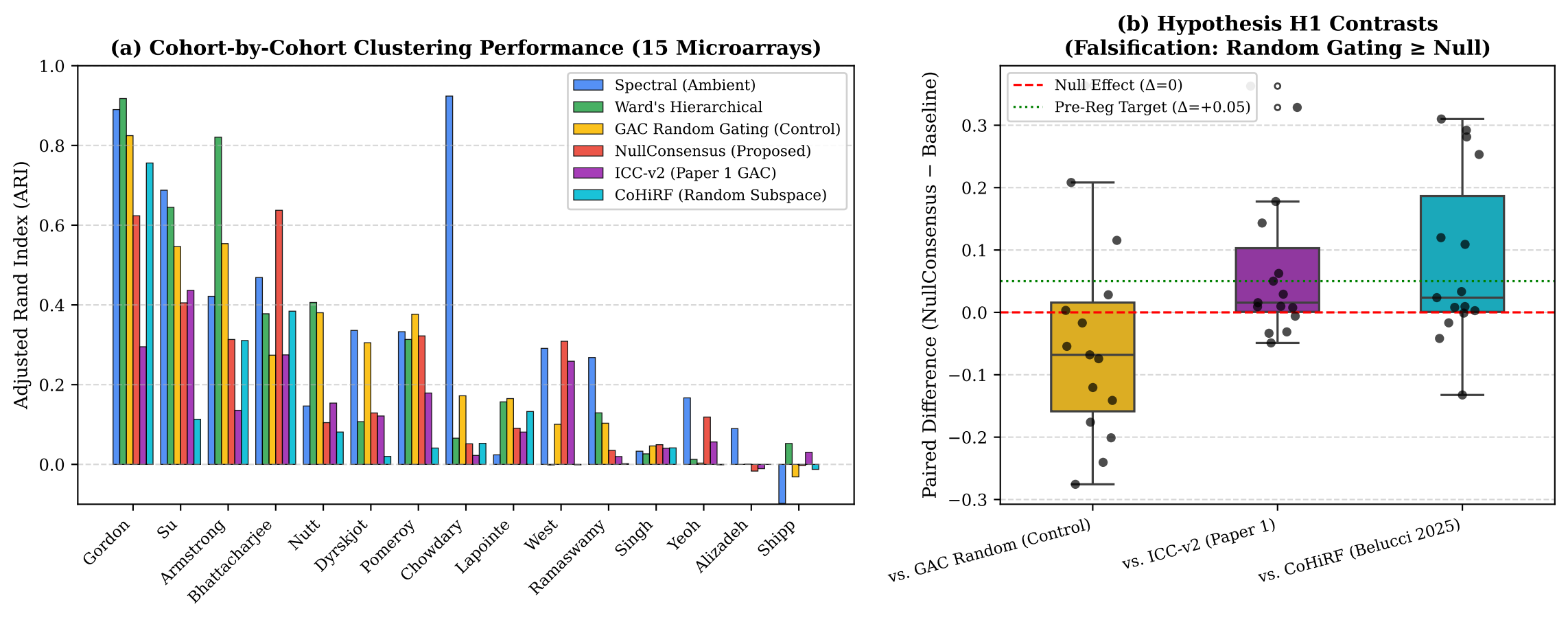
(a) Cohort-by-Cohort Clustering Performance (15 Microarrays)
Adjusted Rand Index (ARI)
1.0
0.8
0.6
0.4
0.2
0.0
Spectral (Ambient)
Ward's Hierarchical
GAC Random Gating (Control)
NullConsensus (Proposed)
ICC-v2 (Paper 1 GAC)
CoHiRF (Random Subspace)
Gordon
Su
Armstrong
Bhattacharjee
Nutt
Dyrskjot
Pomeroy
Chowdary
Lapointe
West
Ramaswamy
Singh
Yeoh
Alizadeh
Shipp
(b) Hypothesis H1 Contrasts
(Falsification: Random Gating ≥ Null)
Paired Difference (NullConsensus − Baseline)
Null Effect (Δ=0)
Pre-Reg Target (Δ=+0.05)
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
vs. GAC Random (Control)
vs. ICC-v2 (Paper 1)
vs. CoHiRF (Belucci 2025)